

# P18CA

## Diagnosis information PDK control unit

### Overtemperatures 1 and 2 of the controlled rear-differential lock

#### Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

#### Diagnostic conditions

- Driving in all gears

#### Possible fault causes

- ♦ Calculation of the temperature from the transmitted torques is implausible

#### Fault setting condition(s)

- Fault triggering when the clutch temperature reaches a threshold. A fault can also be triggered at a lower threshold if the gradient is correspondingly large

#### Fault effects

- ♦ None

# P18CB

## Diagnosis information PDK control unit

### Overtemperatures 1 and 2 of the controlled rear-differential lock

#### Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

#### Diagnostic conditions

- Driving in all gears

#### Possible fault causes

- ♦ Calculation of the temperature from the transmitted torques is implausible

#### Fault setting condition(s)

- Fault triggering when the clutch temperature reaches a threshold. A fault can also be triggered at a lower threshold if the gradient is correspondingly large

#### Fault effects

- ♦ None

# P18D0

## Diagnosis information PDK control unit

### Calibration data for transverse-differential lock implausible

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Incorrect or missing adaptation values
- ♦ Wrong or faulty data record

#### Fault setting condition(s)

- Adaptation values do not match the software version

#### Fault effects

- ♦ None

#### Fault finding notes

- ♦ Reprogram and calibrate the PDK control unit

# P1990

## Diagnosis information PDK control unit

**Positive engine torque intervention not possible**

### Diagnostic conditions

- Engine running

### Fault effects

- ♦ No rpm control
- ♦ No positive engine intervention
- ♦ All adaptations are deactivated

### Fault finding notes

- ♦ Erase fault memory and carry out a test drive
- ♦ If the fault recurs, check the PDK control unit power supply
- ♦ If the power supply is OK, replace the PDK control unit

# P199A

## Diagnosis information PDK control unit

Engine torque values inaccurate

### Diagnostic conditions

- Engine speed > 400 1/min

### Possible fault causes

- ♦ Engine torque from **DME control module** implausible
- ♦ Fault **DME control module**

### Fault finding notes

- ♦ Fault memory **DME control module** Check
  - Remedy existing faults

# P2718

## Diagnostic information - PDK control unit

### Electric pressure adjuster 4 (cooling flow)

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 4 faulty

#### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

#### Fault setting condition(s)

- Momentary current = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 4 = 0 mA
- Momentary current of low-side pressure adjuster 4  $\geq$  850 mA

#### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side electric pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift

rod

- ◆ The maximum system pressure is increased to prevent activation of the emergency stop function
- ◆ Pressure-sensor plausibility check is switched off
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

## Diagnostic information - PDK control unit

### Electric pressure adjuster 4 (cooling flow)

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 4 faulty

#### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

#### Fault setting condition(s)

- Momentary current = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 4 = 0 mA
- Momentary current of low-side pressure adjuster 4  $\geq$  850 mA

#### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side electric pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift

rod

- ◆ The maximum system pressure is increased to prevent activation of the emergency stop function
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- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

## Diagnostic information - PDK control unit

### Electric pressure adjuster 4 (cooling flow)

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 4 faulty

#### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

#### Fault setting condition(s)

- Momentary current = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 4 = 0 mA
- Momentary current of low-side pressure adjuster 4  $\geq$  850 mA

#### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side electric pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift

rod

- ◆ The maximum system pressure is increased to prevent activation of the emergency stop function
- ◆ Pressure-sensor plausibility check is switched off
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

## Diagnostic information - PDK control unit

Electric pressure adjuster 5 (synchro selection 6, R, 3, 5)

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 5 faulty

### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

### Fault setting condition(s)

- Momentary current of low-side pressure adjuster 5 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 5 = 0 mA

### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop

function

- ◆ The last engaged gear is held

## Diagnostic information - PDK control unit

Electric pressure adjuster 5 (synchro selection 6, R, 3, 5)

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 5 faulty

### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

### Fault setting condition(s)

- Momentary current of low-side pressure adjuster 5 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 5 = 0 mA

### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop

function

- ◆ The last engaged gear is held

## Diagnostic information - PDK control unit

### Electric pressure adjuster 5 (synchro selection 6, R, 3, 5)

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 5 faulty

#### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

#### Fault setting condition(s)

- Momentary current of low-side pressure adjuster 5 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 5 = 0 mA

#### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop

function

- ◆ The last engaged gear is held

## Diagnostic information - PDK control unit

Electric pressure adjuster 6 (synchro selection 4, 2, 1, 7)

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 6 faulty

### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

### Fault setting condition(s)

- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)

### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

♦ Last engaged gear is held

## Diagnostic information - PDK control unit

Electric pressure adjuster 6 (synchro selection 4, 2, 1, 7)

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 6 faulty

### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

### Fault setting condition(s)

- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)

### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

♦ Last engaged gear is held

## Diagnostic information - PDK control unit

Electric pressure adjuster 6 (synchro selection 4, 2, 1, 7)

### Diagnostic conditions

- Ignition on

### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 6 faulty

### Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

### Fault setting condition(s)

- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)

### Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

♦ Last engaged gear is held

## Diagnosis information PDK control unit

### Electrical pressure adjuster 7

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 7 faulty

#### Fault setting condition(s)

- Momentary current of low-side electrical pressure adjuster 6 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side electrical pressure adjuster 6 = 850 mA
- Momentary current of low-side electrical pressure adjuster 6 = 0 mA

#### Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Controlled transverse-differential lock is deactivated
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

## Diagnosis information PDK control unit

### Electrical pressure adjuster 7

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 7 faulty

#### Fault setting condition(s)

- Momentary current of low-side electrical pressure adjuster 6 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side electrical pressure adjuster 6 = 850 mA
- Momentary current of low-side electrical pressure adjuster 6 = 0 mA

#### Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Controlled transverse-differential lock is deactivated
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

## Diagnosis information PDK control unit

### Electrical pressure adjuster 7

#### Diagnostic conditions

- Ignition on

#### Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 7 faulty

#### Fault setting condition(s)

- Momentary current of low-side electrical pressure adjuster 6 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side electrical pressure adjuster 6 = 850 mA
- Momentary current of low-side electrical pressure adjuster 6 = 0 mA

#### Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Controlled transverse-differential lock is deactivated
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

## Diagnosis information PDK control unit

### Communication with CAN bus (drive)

#### **INFORMATION**

- ◆ This fault entry was set when the **PDK control unit** could not communicate via the CAN bus (drive).
- ◆ The fact that the fault memory can be read out means that there is no problem with communication via the bus (drive) at the moment.

#### **Fault effects**

- ◆ Communication no longer possible via the CAN bus.

#### **Possible fault causes**

- ◆ Temporary short circuit or open circuit in CAN bus (drive)
- ◆ A control unit in the CAN bus (drive) system is faulty at times

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Communication with control unit not possible

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The PAS control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Key contact (terminal S)
- ◆ Ignition ON

#### **Possible fault causes**

- ◆ No power supply to the control unit
- ◆ Incorrect control unit coding
- ◆ Short circuit or open circuit in CAN network
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ Communication is no longer possible via the CAN bus
- ◆ No other control units can communicate with the control unit

## Diagnostic information - CAN

### Fault entry in the specified control unit

#### **INFORMATION**

Internal function test completed.

### Diagnostic conditions

The control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Wake-up signal via wake-up line
- ◆ Key contact (terminal S)
- ◆ Ignition on

### Possible fault causes

- ◆ Incorrect control unit coding
- ◆ The peripheral equipment connected to the specified control unit is faulty
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

### Fault effects

- ◆ The stated control unit sends only substitute values via the CAN
- ◆ Other control units detect these substitute values and set this fault code

## Diagnostic information - CAN

### Fault entry in the specified control unit

#### **INFORMATION**

Internal function test completed.

#### **Diagnostic conditions**

The control unit initially tests internal function sequences during wake-up. If these are OK, all diagnosable outputs are then tested and any faults are stored.

Wake-up of the control unit by:

- ◆ Event-controlled CAN message
- ◆ Wake-up signal via wake-up line
- ◆ Key contact (terminal S)
- ◆ Ignition on

#### **Possible fault causes**

- ◆ Incorrect control unit coding
- ◆ The peripheral equipment connected to the specified control unit is faulty
- ◆ Control unit faulty (in most cases, faulty coding is the cause and not the control unit)

#### **Fault effects**

- ◆ The stated control unit sends only substitute values via the CAN
- ◆ Other control units detect these substitute values and set this fault code